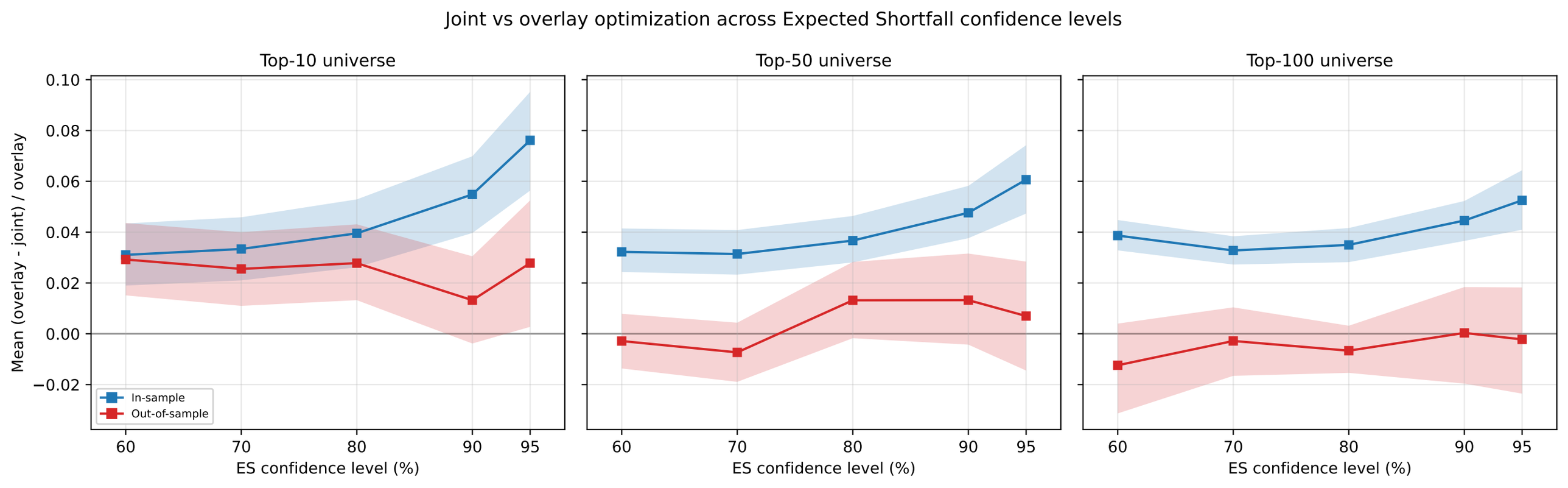
Joint vs overlay optimization across Expected Shortfall confidence levels
Top-10 universe
Top-50 universe
Top-100 universe
Mean (overlay - joint) / overlay
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
In-sample
Out-of-sample
60
70
80
90
95
ES confidence level (%)